

WORK EXPERIENCE

- **Amazon** August 2022 - June 2026
 - *Software Development Engineer II*
 - Designed and implemented a flexible guardrail layer for Amazon’s first generative AI customer support chatbot, integrating in-house ML models and Amazon Bedrock APIs through a unified interface. Collaborated directly with internal science teams to drive API contract alignment and integrated their models to validate customer inputs and model outputs.
 - Designed and built automated ML infrastructure using Amazon SageMaker and CI/CD pipelines to streamline model training, testing, and production deployment. Integrated a real-time issue prediction model into the customer service chatbot, reducing average customer interaction time by 9% (440s to 400s).
 - Extended Prime Video’s automated returns experience from the US to the UK and Japan, enabling eligible customers to self-service returns for otherwise non-refundable orders through Amazon’s customer service chatbot without contacting a customer service agent.
 - Redesigned the backend architecture powering device selection in Amazon’s customer service chatbot, resolving reliability issues and ensuring registered devices and their images are consistently displayed.
 - Owned and implemented the integration of Amazon Demand-Side Platform (ADSP) into the Retailer Ad Service’s attribution service, extending existing Sponsored Products (SP) attribution to also support display and video advertising. Built unified data pipelines to ingest upstream SP and ADSP data, and developed attribution logic in Spark SQL within AWS Clean Rooms to jointly perform attribution across both programs.
- **Agency for Science, Technology and Research (A*STAR)** Jun 2018 - Sept 2018
 - *Data Science Intern*
 - Developed **SeizNet**, a CNN-based seizure detection model, achieving 95.8% sensitivity with 0.17 false alarms/hour on EEG data from 29 patients; demonstrated that a 2-channel configuration achieved 93.3% sensitivity, outperforming a traditional SVM baseline using the full set of EEG channels.

TECHNICAL SKILLS

Programming Languages: Python, Java, TypeScript, Scala, JavaScript, SQL, C++**Distributed Systems & Data:** Spark, Glue, EMR, DynamoDB, SQS/SNS, AWS Clean Rooms**AI & MLOps:** SageMaker, Bedrock, Large Language Models (LLMs), Keras, TensorFlow**Cloud & Infrastructure:** AWS (CDK, CloudFormation, Lambda, S3), Docker, Linux, CI/CDEDUCATION

- **University of Southern California** May 2022
 - *Master of Science in Computer Science, CGPA: 3.96/4.00*
 - Selected Courses: Machine Learning, Analysis of Algorithms, Database Systems, Applied Machine Learning for Games, Estimation Theory, Random Processes, Applied Natural Language Processing
- **Bilkent University** Jun 2019
 - *Bachelor of Science in Electrical and Electronics Engineering, CGPA: 3.93/4.00*

HONORS & AWARDS

- **Two-time Amazon org-wide hackathon winner among 20+ competing teams**
- **Annenberg Fellowship with admission to University of Southern California (USC) PhD program, 2019**
- **IEEEExtreme Programming Competition**
 - **IEEEExtreme 10.0, 2016:** Ranked 2nd in Turkey, 20th in IEEE Region 8, and 65th overall
 - **IEEEExtreme 9.0, 2015:** Ranked 3rd in Turkey, 30th in IEEE Region 8, and 82nd overall
- **Nationwide University Entrance Exam (LYS):** Ranked in top 0.01% among 2 million students in Turkey, 2014

PUBLICATIONS

M. T. Avcu, Z. Zhang, and DWS. Chan “Seizure Detection Using Least EEG Channels by Deep Convolutional Neural Networks” IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2019